

Exponential Shot-Noise Ornstein–Uhlenbeck Dynamics: Exact Semigroup, Gamma Equilibrium, and Polynomial Filtrations

Route-A source-local certificate HCS-C335

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Abstract

We close the positive Ornstein–Uhlenbeck process driven by Poisson arrivals with exponential marks. Its all-time transition Laplace transform follows directly from the marked-Poisson functional. The same formula yields the unique Gamma equilibrium, sharp Wasserstein contraction, all stationary moments and cumulants, and the exact covariance.

1 Frozen process and semigroup

Fix $\gamma, \kappa, \beta > 0$. Let N_t be Poisson of rate κ , let T_j be its jump times, and let the independent marks Y_j have exponential rate β . On $[0, \infty)$ consider

$$dX_t = -\gamma X_t dt + dJ_t, \quad J_t = \sum_{T_j \leq t} Y_j, \quad X_0 = x \geq 0. \quad (1)$$

Theorem 1 (Exact affine Markov closure). *For $t, s \geq 0$, with $\alpha = \kappa/\gamma$,*

$$X_t = e^{-\gamma t} x + \sum_{T_j \leq t} e^{-\gamma(t-T_j)} Y_j, \quad (2)$$

$$\mathbb{E}_x e^{-sX_t} = e^{-se^{-\gamma t} x} \left(\frac{\beta + se^{-\gamma t}}{\beta + s} \right)^\alpha. \quad (3)$$

The unique invariant probability is Gamma with shape α and rate β . For every $p \geq 1$,

$$W_p(P_t(x, \cdot), P_t(y, \cdot)) = e^{-\gamma t} |x - y|. \quad (4)$$

In stationarity,

$$\mathbb{E} X^n = \frac{(\alpha)_n}{\beta^n}, \quad \text{cum}_n(X) = \frac{\alpha(n-1)!}{\beta^n}, \quad \text{Cov}(X_t, X_0) = \frac{\alpha}{\beta^2} e^{-\gamma|t|}. \quad (5)$$

Finally, the generator preserves $\mathcal{P}_m = \text{span}\{1, x, \dots, x^m\}$, and its restriction has exactly the simple eigenvalues $0, -\gamma, \dots, -m\gamma$. This last assertion concerns each fixed finite-dimensional filtration only; it is not a statement about the full L^2 spectrum, completeness, normality, or reversibility.

Proof of the path and transform. Variation of constants between successive jumps proves (2). The Laplace functional of a marked Poisson process gives

$$\begin{aligned} \log \mathbb{E} \exp \left[-s \sum_{T_j \leq t} e^{-\gamma(t-T_j)} Y_j \right] &= \kappa \int_0^t \left(\frac{\beta}{\beta + se^{-\gamma u}} - 1 \right) du \\ &= \frac{\kappa}{\gamma} \log \frac{\beta + se^{-\gamma t}}{\beta + s}. \end{aligned}$$

Multiplication by the deterministic initial term proves (3). Factoring the right side first at decay $r_1 = e^{-\gamma t_1}$ and then at $r_2 = e^{-\gamma t_2}$ cancels the intermediate denominator and leaves $r_1 r_2$, proving the Chapman–Kolmogorov law directly. $\square$

The phrase *shot-noise Laplace semigroup owner* is the round-zero certificate. Formula (3), not a simulation, owns the transition law.

2 Equilibrium, contraction, and correlations

Sending $t \rightarrow \infty$ in (3) gives

$$\left(\frac{\beta}{\beta+s}\right)^\alpha,$$

the $\text{Gamma}(\alpha, \text{rate } \beta)$ transform. Substitution into the semigroup formula proves invariance. If two copies use the same marked Poisson process, (2) gives the deterministic difference $e^{-\gamma t}(x - y)$. Their laws are translates, so this coupling gives the upper bound in (4); Jensen's inequality applied to their mean difference gives the same lower bound. Any invariant law must, after its Laplace transform is propagated and $t \rightarrow \infty$, have the Gamma transform. This proves uniqueness without presupposing a moment.

Expansion of the logarithm of the Gamma transform yields the cumulants in (5); differentiation gives its rising-factorial moments. The conditional mean is

$$\mathbb{E}[X_t \mid X_0] = e^{-\gamma t} X_0 + \frac{\kappa}{\gamma\beta}(1 - e^{-\gamma t}).$$

After stationary centering, multiplication by X_0 gives the covariance for $t \geq 0$. Scalar stationarity supplies the displayed even extension. These identities also agree with the classical shot-noise and Lévy-driven OU sources [1, 2, 3].

References

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